

Lagrangian of the Fine-Structure Constant: Analytical Derivation of Quantum Gravity

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Abstract

We present a unified Lagrangian formulation in which electromagnetism and gravity emerge from complementary projections of a single 5-dimensional Projected Entangled Pair State (PEPS). The fundamental cell is a Planck sphere of radius l_p , and its informational volume is axiomatically defined as $I_s = \pi/6$ from spinorial holography.

The complete unified Lagrangian is the sum of three terms:

$$\mathcal{L}_{\text{unified}} = \mathcal{L}_{\text{volume}} + \mathcal{L}_{\text{EM}} + \mathcal{L}_{\text{hinge}},$$

where:

- $\mathcal{L}_{\text{volume}} = \frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi - \frac{\mu^2}{2}(\Phi^2 - (\pi^4 + 1))^2$ is the tesseract volume potential (gravitational sector);
- $\mathcal{L}_{\text{EM}} = \sqrt{-g(t)}\left(\dot{d}^2 - \frac{1}{64}(\nabla d)^2\right)$ is the Polyakov string action with tension $T = 8$ (electromagnetic sector, Alpha Series, concept DOI: [10.5281/zenodo.19802606](https://doi.org/10.5281/zenodo.19802606));
- $\mathcal{L}_{\text{hinge}} = \Phi^2\left|e^{-i\pi/6}\Psi_d - \frac{4}{3\pi}\Psi_\Phi\right|^2$ is the hinge term that locks the string state Ψ_d to the gravitational volume state Ψ_Φ at the minimal spinorial resolution angle $\chi = \pi/6$.

The hinge term $\mathcal{L}_{\text{hinge}}$ is the mathematical suture that unifies the two sectors. It does not stand alone; it couples $\mathcal{L}_{\text{volume}}$ and $\mathcal{L}_{\text{EM}}$ into a single coherent structure. Solving the combined action on-shell yields the bare gravitational coupling:

$$\ln\left(\frac{G_{\text{bare}}m_e^2}{\hbar c}\right) = -\frac{\pi}{3}(\pi^4 + 1).$$

No free parameters are introduced. The fine-structure constant α (derived in the Alpha Series) and the bare gravitational coupling α_G (derived here) are two eigenvalues of the same topological operator, distinguished only by the dimensionality of

their projection: 1D for electromagnetism (circular MPS, bond dimension $D = 45$), 5D for gravity (PEPS, Planck sphere cell). The construction is closed, falsifiable, and requires no external parameters beyond π , the string tension $T = 8$, and the overlap angle $\pi/6$.

Keywords: Lagrangian, quantum gravity, GUT, grand unified theory, PEPS, MPS, tensor networks, topological invariants, fine-structure constant, Alpha Series, Polyakov string, Planck sphere, holographic principle, spinorial projection, tesseract volume, bare gravitational coupling, dimensional reduction, lattice quantum field theory, Monte Carlo simulation, annealing, symmetry breaking, hinge mechanism, topological locking, 5D field theory, geometric unification, no free parameters, falsifiable theory.

Alpha Series concept DOI: [10.5281/zenodo.19802606](https://doi.org/10.5281/zenodo.19802606)

Contents

1	Introduction	3
2	Geometric Preliminaries	3
2.1	Informational Volume of the Planck Cell	3
2.2	The Overlap Angle	4
2.3	The String Tension and Its Dual Role	4
3	The Lagrangian of a Single PEPS Node	4
3.1	Definition of the Lagrangian	4
3.2	Definition of Fields	4
3.3	Interpretation	5
4	Derivation of the Bare Gravitational Coupling	5
4.1	Step 1: Vacuum saturation	5
4.2	Step 2: Reduction of the hinge term on-shell	5
4.3	Step 3: Angular integration and effective action	6
4.4	Step 4: Spinorial doubling of the angle	6
4.5	Step 5: Evaluation of the effective action	6
4.6	Step 6: Theorem	6
4.7	Step 7: Relation to the measured gravitational constant	7
5	Topological Invariant	7
6	Numerical Verification	7
7	Conclusions	7

1 Introduction

Previous work (concept DOI: [10.5281/zenodo.19802606](https://doi.org/10.5281/zenodo.19802606)) established a geometric derivation of the fine-structure constant from a circular Matrix Product State (MPS) with bond dimension $D = 45$, using only π , the string tension $T = 8$, and the 24 transverse modes of the bosonic string. No free parameters were introduced. The master formula from that derivation is

$$\alpha^{-1} = \ln \lambda_{\max} - \pi,$$

where $\lambda_{\max}$ is the largest eigenvalue of the MPS transfer matrix, converging numerically to $\alpha^{-1} = 137.035999177$ in agreement with CODATA 2022 [2].

In this work we extend the same geometric principles to the gravitational sector. We construct a 5-dimensional Projected Entangled Pair State (PEPS) [4] whose elementary cell is a **Planck sphere** of radius l_p . The isotropy of the vacuum forces the informational volume of the cell to be determined by a minimal informational radius.

The local Lagrangian of the PEPS is built from three structurally determined terms. Solving it on-shell yields an exact expression for the bare gravitational constant, with no free parameters. The derivation is analytical, self-contained, and every mathematical step is explicitly justified.

The two sectors — electromagnetic and gravitational — are not independent. They share the same fundamental constants: π , the string tension $T = 8$, and the overlap angle $\chi = \pi/6$. Their unification is not an additional postulate but a structural necessity: both derivations describe complementary projections of a single entangled tensor network. The Alpha Series captures the 1D projection (the string’s vibrational spectrum), while the present 5D PEPS captures the 5D projection (the lattice’s volumetric response).

A companion Python code, `PEPS_5D_NODE_SIMULATOR.QuantumGravity.py`, is provided as supplementary material. It implements three numerical verifications: (1) spontaneous symmetry breaking to $\langle \Phi^2 \rangle = \pi^4 + 1$ via simulated annealing; (2) deterministic on-shell hinge action integration to $S_{\text{eff}} = \frac{\pi}{3}(\pi^4 + 1)$; (3) numerical verification of the topological invariant $d^5 F/dL^5 = -40$. All simulations confirm the analytical derivations to machine precision.

2 Geometric Preliminaries

2.1 Informational Volume of the Planck Cell

Axiom 2.1 (Informational Volume). *The fundamental cell of spacetime is a sphere of Planck length radius l_p . The intersection between the 5D holographic screen of the node and its 3D spinorial projection determines a net informational volume*

$$I_s = \frac{\pi}{6}.$$

This value is motivated by the fermionic nature of the minimal resolution angle $\pi/6$ (the angle required for a spinor to return to its initial state up to a sign after projection from 5D to 3D). The axiom is taken as foundational.

2.2 The Overlap Angle

In a cosmological FLRW metric, the spatial section is a 3-sphere. The minimal angle that allows consistent embedding of a 1D closed string into the 3-sphere, given the spinorial holonomy, is

$$\chi = \frac{\pi}{6}.$$

This is the **overlap angle** between the 5D vacuum and its 3D projection. It appears in the hinge term of the Lagrangian and, due to the spinorial nature of matter, doubles to $\pi/3$ in the effective action.

2.3 The String Tension and Its Dual Role

The string tension $T = 8$ appears identically in both sectors:

- In the Alpha Series, it determines the fine-structure constant through the MPS transfer matrix spectrum.
- In the present PEPS construction, it appears as the factor $1/64 = 1/T^2$ in the string term $\mathcal{L}_{\text{EM}} = \sqrt{-g(t)}(\dot{d}^2 - \frac{1}{64}(\nabla d)^2)$.

This dual role is not a coincidence: the same Polyakov string [3] that generates the electromagnetic coupling is woven into the fabric of the 5D PEPS lattice.

3 The Lagrangian of a Single PEPS Node

The total action is a sum over all nodes:

$$S_{\text{tot}} = \sum_j \int d\tau \mathcal{L}_{5D}^{(j)}.$$

3.1 Definition of the Lagrangian

$$\boxed{\begin{aligned} \mathcal{L}_{5D} = & \frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi - \frac{\mu^2}{2} \left(\Phi^2 - (\pi^4 + 1) \right)^2 \\ & + \sqrt{-g(t)} \left(\dot{d}^2 - \frac{1}{64} (\nabla d)^2 \right) \\ & + \Phi^2 \left| \exp\left(-i\frac{\pi}{6}\right) \Psi_d - \frac{4}{3\pi} \Psi_\Phi \right|^2 \end{aligned}}$$

3.2 Definition of Fields

- $\Phi(x^\mu)$: real scalar field representing the volume of the 4D tesseract. Its vacuum expectation value will be $\Phi_{\text{VEV}}^2 = \pi^4 + 1$.
- $d(x^\mu)$: scalar displacement field of the electromagnetic string. The term $\dot{d}^2 - \frac{1}{64}(\nabla d)^2$ is the worldsheet action with tension $T = 8$ (since $1/T^2 = 1/64$).

- $\Psi_d(x^\mu)$: complex scalar field representing the quantum state of the string. This field is the direct link to the Alpha Series, where its fluctuations generate the fine-structure constant.
- $\Psi_\Phi(x^\mu)$: complex scalar field representing the quantum state of the tesseract volume. Its amplitude is topologically locked to the inverse of the 3-sphere spinorial mapping invariant $I_0 = 4/(3\pi)$ to ensure gauge consistency:

$$|\Psi_\Phi| = \frac{3\pi}{4}, \quad \Psi_\Phi = \frac{3\pi}{4}e^{i\theta},$$

where θ is the angular coordinate on the fifth-dimensional compactification circle.

- $\sqrt{-g(t)}$: determinant of the FLRW metric.

The amplitude $3\pi/4$ is the inverse of $I_0 = 4/(3\pi)$, i.e., I_0^{-1} . This topological locking guarantees that the hinge term collapses to a pure geometric measure without residual constants.

3.3 Interpretation

- **Volume term:** The potential has minima at $\Phi_{\text{VEV}}^2 = \pi^4 + 1$. The number π^4 is the volume of a 4D hypercube of side π ; the $+1$ is the topological unit (Euler characteristic of the base point).
- **String term (EM):** Describes the electromagnetic string with tension $T = 8$ [3] in expanding spacetime. This term alone, when quantized on a circle, yields the fine-structure constant of the Alpha Series.
- **Hinge term:** Contains the overlap angle $\pi/6$ and the invariant $I_0 = 4/(3\pi)$, which emerges from the normalized topological trace of the 3-sphere embedding. The prefactor Φ^2 ensures the correct scaling. The field Ψ_Φ has amplitude I_0^{-1} to cancel the I_0 factor in the coupling, leaving a pure phase. This term is the mathematical suture that locks the string state Ψ_d to the gravitational volume state Ψ_Φ at the minimal spinorial resolution angle.

4 Derivation of the Bare Gravitational Coupling

4.1 Step 1: Vacuum saturation

On-shell, kinetic fluctuations vanish and the potential must be zero:

$$V(\Phi) = \frac{\mu^2}{2} (\Phi^2 - (\pi^4 + 1))^2 = 0 \quad \Rightarrow \quad \Phi_{\text{VEV}}^2 = \pi^4 + 1.$$

4.2 Step 2: Reduction of the hinge term on-shell

On-shell, string fluctuations vanish: $\Psi_d = 0$. The field Ψ_Φ assumes its vacuum configuration:

$$\Psi_\Phi = \frac{3\pi}{4}e^{i\theta}.$$

Substituting into the hinge term:

$$\mathcal{L}_{\text{hinge}}^{(\text{on-shell})} = \Phi_{\text{VEV}}^2 \left| 0 - \frac{4}{3\pi} \cdot \frac{3\pi}{4} e^{i\theta} \right|^2 = \Phi_{\text{VEV}}^2 |-e^{i\theta}|^2 = \Phi_{\text{VEV}}^2.$$

The cancellation is exact. No residual constants remain. The hinge Lagrangian collapses to Φ_{VEV}^2 , a pure geometric density.

4.3 Step 3: Angular integration and effective action

The Euclidean action is obtained by integrating over the 5D volume. The integration over the four non-topological dimensions (3 space + time) cancels between numerator and denominator of the partition function ratio. The remaining integration over the fifth-dimensional angle θ yields:

$$S_{\text{eff}} = \int_0^{\Delta\theta} \Phi_{\text{VEV}}^2 d\theta,$$

where $\Delta\theta$ is the effective angular interval traced by the hinge.

4.4 Step 4: Spinorial doubling of the angle

The hinge term contains $\exp(-i\pi/6)$. Because matter is spinorial, a rotation of $\theta/2$ on the fermionic state corresponds to a geometric rotation θ on the bosonic volume. Hence:

$$\Delta\theta = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3}.$$

4.5 Step 5: Evaluation of the effective action

$$S_{\text{eff}} = \int_0^{\pi/3} (\pi^4 + 1) d\theta = \frac{\pi}{3} (\pi^4 + 1).$$

4.6 Step 6: Theorem

Theorem 4.1 (Bare Gravitational Coupling). *Given the 5D PEPS node with vacuum expectation value $\Phi_{\text{VEV}}^2 = \pi^4 + 1$ and a spinorial overlap angle $\Delta\theta = \pi/3$, the bare gravitational coupling is:*

$$\ln \left(\frac{G_{\text{bare}} m_e^2}{\hbar c} \right) = -\frac{\pi}{3} (\pi^4 + 1).$$

Proof. In lattice quantum field theory [5], the amplitude for the topological state is $\exp(-S_{\text{eff}})$. Thus

$$\alpha_G^{(\text{bare})} = \exp \left(-\frac{\pi}{3} (\pi^4 + 1) \right).$$

Taking the logarithm yields the master formula. □

No free parameters appear. The topological locking condition $|\Psi_\Phi| = I_0^{-1}$ guarantees the exact cancellation of all geometric constants.

4.7 Step 7: Relation to the measured gravitational constant

The coupling $\alpha_G^{(\text{bare})}$ derived above is the bare gravitational constant evaluated at the Planck scale, specifically at the scale of the tesseract volume $\pi^4 + 1$. The measured value G_{CODATA} is the effective coupling at cosmological scales, renormalized by the integration over vacuum fluctuations from the Planck scale to the Hubble horizon. The exact scaling function is governed by the gravitational renormalization group flow from the cutoff scale to the infrared cosmological horizon, which will be determined in a subsequent work. The present derivation provides the bare anchoring value at the cutoff scale.

5 Topological Invariant

The effective action as a function of the tesseract side length L (with $L = \pi$ at the physical scale) is:

$$F(L) = -\frac{L}{3}(L^4 + 1) = -\frac{L^5}{3} - \frac{L}{3}.$$

The fifth derivative extracts the coefficient of the highest-degree term, eliminating lower-order fluctuations:

$$\frac{d^5 F}{dL^5} = -\frac{5!}{3} = -40.$$

The choice of the fifth derivative is dictated by the 5D nature of the construction: in a d -dimensional topological field theory, the d -th derivative of the action with respect to the geometric scale yields a pure integer invariant. Here $d = 5$, hence the fifth derivative. The constant -40 is the topological signature of the 5D PEPS architecture.

6 Numerical Verification

The analytical derivations are confirmed by three independent numerical simulations implemented in the accompanying Python code `PEPS_5D_NODE_SIMULATOR_Quantum_Gravity.py`:

1. **VEV Saturation:** Simulated annealing of the double-well potential $V(\Phi) = \frac{\mu^2}{2}(\Phi^2 - (\pi^4 + 1))^2$ demonstrates spontaneous symmetry breaking to $\langle \Phi^2 \rangle = \pi^4 + 1$ with relative error $< 10^{-9}$.
2. **Hinge Action Collapse:** Deterministic numerical integration of the on-shell hinge Lagrangian over $\theta \in [0, \pi/3]$ yields $S_{\text{eff}} = \frac{\pi}{3}(\pi^4 + 1)$ to machine precision (error $< 10^{-15}$).
3. **Topological Invariant:** Finite-difference computation of $d^5 F/dL^5$ with step size $h = 0.5$ returns -40 exactly, with numerical fluctuations at the 10^{-12} level.

All simulations are reproducible and require no free parameters.

7 Conclusions

We have derived the bare gravitational constant from a 5D PEPS with Planck sphere cell. The derivation uses only geometric and topological inputs:

- Informational volume axiom: $I_s = \pi/6$.
- Overlap angle: $\chi = \pi/6$, doubling to $\Delta\theta = \pi/3$.
- Tesseract volume: $\Phi_{\text{VEV}}^2 = \pi^4 + 1$.
- Topological locking: $|\Psi_\Phi| = I_0^{-1} = 3\pi/4$ with $I_0 = 4/(3\pi)$ from the normalized topological trace of the 3-sphere.

The derivation is a forced cascade:

$$\Phi_{\text{VEV}}^2 = \pi^4 + 1 \longrightarrow \Delta\theta = \pi/3 \longrightarrow S_{\text{eff}} = \frac{\pi}{3}(\pi^4 + 1) \longrightarrow \ln \alpha_G^{(\text{bare})} = -\frac{\pi}{3}(\pi^4 + 1).$$

The fifth derivative -40 is a topological invariant confirming the 5D rigidity. No free parameters are introduced.

Unification with the Electromagnetic Sector

The 5D PEPS construction presented here serves as the geometric host for the electromagnetic string of the Alpha Series [1]. The convergence of the two sectors under the same topological invariants reveals a unified architectural origin:

- The string tension $T = 8$ appears identically in both the Alpha Series (where it determines the fine-structure constant $\alpha^{-1} = \ln \lambda_{\text{max}} - \pi$) and in the PEPS Lagrangian (where it determines the factor $1/64 = 1/T^2$ in $\mathcal{L}_{\text{EM}}$).
- The overlap angle $\chi = \pi/6$ governs both the spinorial projection of the 5D vacuum and the phase interference in the hinge term that couples the string state Ψ_d to the gravitational volume state Ψ_Φ .
- The same geometric constant $\pi^4 + 1$ that anchors the tesseract volume also appears as the vacuum expectation value that saturates the double-well potential.

Thus, the fine-structure constant α emerges from the internal dynamics of the closed string in a circular MPS projection, while the bare gravitational coupling α_G emerges from the resistance that the 5D PEPS lattice imposes on that same string when stretched between nodes. The hinge term $\Phi^2 |e^{-i\pi/6} \Psi_d - I_0 \Psi_\Phi|^2$ is the mathematical suture: it forces the string and the volume to lock at the minimal spinorial resolution angle $\pi/6$, preventing the two sectors from decoupling.

The present construction suggests that gauge forces and gravity arise from different projections of the same underlying quantum geometric state. The Alpha Series captures the 1D projection (the string's vibrational spectrum), while the 5D PEPS captures the 5D projection (the lattice's volumetric response). Their unification is not an additional postulate but a structural necessity: both derivations share the same constants (π , $T = 8$, $\pi/6$) because they describe complementary aspects of a single entangled tensor network.

The construction is closed, falsifiable, and requires no free parameters.

Data Availability

The Python simulation code `PEPS_5D_NODE_SIMULATOR_Quantum_Gravity.py` is available at:

<https://github.com/massimilianoblandino/PEPS-5D-Quantum-Gravity>

All numerical results presented in this paper can be reproduced by running the provided script.

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